

Data systems and decision-making: State ID/DD agencies and their employment data collection systems

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INTRODUCTION

The growing emphasis on government accountability at the state and federal levels has increased interest in the collection and use of outcome data. Moreover, research has found that high performing states in integrated employment generally have a clear and visible data collection system that includes individual outcome data (Hall, Butterworth, Winsor, Gilmore, & Metzel, 2007). But what are the most important elements in designing and using a system? Stakeholders have raised questions regarding how to develop and implement an effective data collection system, which variables provide the most utility for influencing policy, and how to use data as a strategic planning tool.

This topical chapter offers a qualitative, in-depth look into how employment data is collected and used throughout multiple levels within state ID/DD systems. This chapter details findings from state-level case studies conducted in four states: Florida, Massachusetts, New Hampshire, and Washington. Case study research focused specifically on the development and use of employment data collection systems and their relationship to priorities, decision-making, and policy development.

Understanding ID/DD data systems

To put this chapter in context, it is useful to understand the national scope of ID/DD data systems as they relate to employment. As a part of the FY2007 National Survey of Day and Employment Programs, data were requested from state intellectual disability and developmental disability (ID/DD) agencies regarding the sources of information used to report the total number of individuals served in the following services categories: integrated employment, facility-based work, community-based non-work, and facility-based non-work. Data on the source of information is an important factor to note when comparing each state's service outcomes over time (Bhattarai & Winsor 2008). Information on the source of the data can help to explain unexpected trends in state service distribution when the state has not implemented changes in policy or practice. Ensuring that the source of the data used by each state is consistent over time can improve the reliability of longitudinal data. States were given the six choices listed in the table below and asked to identify whether they used each source to complete the service category data and which choice was the primary source of their data. Thirty-seven states responded to the first question. Sixteen of those states used more than one source. However when asked which source was the primary source of data, the majority of states reported that service funding records were their primary source.

Table 1. Sources of data used for reporting outcomes

Data Source	Number of states who reported that they used this source (n=37)	Number of states who reported using this as their primary source (n=37)
Service funding records	26	19

(e.g. number receiving funding by service)		
Data collected at the provider level on services provided	17	12
Other sources	8	3
Data collected at the individual level on employment outcomes	7	3
Case management records	6	0
The number of individuals designated for service in provider contracts	4	0

In a secondary question states were asked if they collect data at the individual level on employment outcomes; twenty states reported they collect data at the individual level on employment outcomes. Those states were asked a series of follow-up questions related to data collection on individual integrated employment placements that can be used to assess the quality of the employment outcomes. Of these 20 states, 18 reported the type of data they collect. The most commonly collected data elements were type of job, wages earned, and hours worked (See Table 2).

While there was no correlation between the percentage of individuals a state supported in integrated employment and the specific type of outcome data they collect, 75 percent of states who supported a high percentage (more than 40 percent) of individuals in integrated employment in FY2007 reported collecting information on at least three of the data types. Only 33 percent of states that supported a moderate percentage (between 20 percent and 40 percent) of individuals in integrated employment reported collecting this same number of data types; and for states supporting a low percentage (less than 20 percent) of individuals in integrated employment only 28.5 percent of states collected data on at least three of the data types. This finding is consistent with that of Hall, Butterworth, Winsor, Gilmore, and Metzel (2007), which concluded that ID/DD agencies that consistently produce high rates of integrated employment have implemented statewide outcome data systems to assess the quality of their employment outcomes.

Table 2. Employment outcomes data elements collected at the individual level

Data type	Number of states collecting this data (n=18)
Type of jobs (e.g., individual, group supported employment)	16
Wages	16
Hours worked	16
Job tenure/longevity	10
Occupation or industry a person is employed in	10
Source of wages (e.g., employer or	9

provider)	
Benefits received from employer	8

METHODOLOGY

We interviewed stakeholders in 2008 from four states to discuss the strategies, successes and challenges within their employment data collection systems and reviewed state documents related to their data collection systems.

Sample selection. States were selected using a purposeful two-phase sampling strategy that ensured representation across key elements, placing particular emphasis on the design, use, and integration of the state ID/DD agencies' employment data collection system. In the first phase, a comprehensive document review process was undertaken and included several states which were known to have some type of employment data collection system in place. This knowledge came from past ICI case study research, involvement of some of the states in the State Employment Leadership Network, and ongoing involvement with ICI's National Survey on Day and Employment Services. These ongoing activities offered enough evidence to be able to list and sort states according to important data-related variables. These states included CT, DE, FL, MA, NH, WV, WI, CO, CA, LA, WA, OR, AZ, NM, TX, and SC. Data from these states were organized into a matrix which sorted data by the following categories: frequency of data collection; population targets for data collection; data elements collected; who is providing the data; how and to what other systems the data is tied; and how data is used and shared across the state. This matrix helped to compare the states across similar variables and allowed the researchers to engage in the second phase of sampling, which involved choosing four of these states that represented a range of data systems and models for use of the data.

Data collection. Researchers used document review and semi-structured interviews to better understand states' employment data collection systems. First, research staff conducted a review of literature and existing documentation related to data collection systems and strategies. The information obtained was organized within the matrix that provided a platform from which to choose the case study states.

Four states were chosen to participate in more in-depth follow-up interviews. These states were MA, FL, WA, and NH. These states were chosen because they had engaged in strategic efforts to create well-developed data systems to capture integrated employment outcomes, because they offered different approaches to data collection, management and use and because of the lessons they could provide to other states in earlier stages of data system development. In each state, ID/DD central office and regional staff participated in hour-long tape recorded telephone conversations about their data systems. Researchers also interviewed staff from community rehabilitation providers (CRPs) in each state to ensure diverse opinions and perspectives were represented. Interviews with CRP staff focused on their experiences in providing data to their state's ID/DD agency, and their perceptions on how the data is used. In total, 17 individuals participated in interviews.

Data analysis. Thematic analysis techniques were used to review and analyze the literature, documentation, and data collected, resulting in the development of a matrix on state data collection policies and methodologies. Interview data was analyzed using a qualitative data

analysis approach. Individual descriptive case studies of each state's funding system can be found elsewhere (e.g. Hall, Winsor, Butterworth, 2009, Winsor, Butterworth, Hall, 2009).

FINDINGS

The findings section discusses both concrete strategies that states used in developing their systems, as well as thematic elements that were present across states. The section concludes with a review of the challenges and lessons learned from states as they moved forward in the evolution of their data systems.

Data systems: A reflection of new and ongoing priorities

The core reason these states have implemented and supported data collection systems is to emphasize priorities. States are paying attention to outcome data because they understand that knowing how many people are working is the key to moving an employment agenda forward. Being able to track how many people are both currently working and entering community employment helps states to develop and then understand their progress toward goals. The continued collection of employment data helps to keep the spotlight on employment in a sea of other priorities with which a state ID/DD agency must contend.

Some of the states studied have understood the link between data and priorities for several years, while other systems have only recently begun to emphasize the data collection process. Florida, for instance, developed its data collection system to track the outcomes of its 5 Year Employment Initiative, begun in 2004. Florida's SETS (Supported Employment Tracking System) was developed to provide a comprehensive and accurate picture of the state's progress in reaching its goals around employment. Prior to 2004, the state had a data collection system in place but it was less automated and standardized. The new database that is now in place reflects a more defined commitment to accuracy and buy-in from all those involved with employment services.

In 2002, the Massachusetts ID/DD developed a contractual requirement that employment services provider performance be tracked through outcome measures. A new Request for Responses (RFR) for Employment Support Services emphasized robust reporting requirements, and consequently the ID/DD agency shaped its employment data collection system to focus on what it viewed as key outcomes for measuring success around employment. Overall its intrinsic commitment to greater community employment supported the development of an employment data system that focused on individual outcomes.

States like New Hampshire and Washington have been comprehensively collecting employment data for a much longer time. New Hampshire began collecting data in 1994 and has built on lessons learned to make its current system standardized and efficient. The state started with a simple question, "How many people are working?" and evolved the data collection effort to meet the changing needs of its system. The data collection system has continued to allow New Hampshire to keep employment at the forefront of its service delivery goals. Washington's system is linked to its billing system, emphasizing the state's long-standing focus on employment outcomes as a systemic priority. The collection of billing and reporting data on individual outcomes is an additional method to ensure that providers are fulfilling their

obligation to support individuals in community employment or in services that support the individual's employment plan.

Data collection strategies

Each state has a structure in place that attempts to standardize how data is collected. However, in some states the details of actually obtaining the data varies by regional office, which was noted by respondents to sometimes have either positive or negative implications.

Florida. In each of Florida's 14 regions, there is a supported employment (SE) liaison that is responsible for collecting employment data. On a monthly basis, each SE liaison documents the outcomes of individuals within his or her region. This is done through outreach to supported employment service providers and Support Coordinators in the region who are supporting individuals who are employed and receiving services from the state ID/DD agency. The way in which the data is reported at this level varies. Depending on the region and the provider, some providers mail their data, while others use email. In one region, each provider develops their own form based on the data that the SE liaison requests. Providers submit the data to the liaison, and he enters it into a hardcopy book and then into the electronic SETS data systems. In another region, the SE liaison works with her IT staff to get a point-in-time picture of the data at the end of each month, and she works with providers, coordinators, and families to get an accurate update of this data each month. Once she gets the data, she enters it into the SETS database. Additionally, each individual file maintains a completed census form which includes demographic data that does not change month to month.

Massachusetts. In Massachusetts, data are collected once per year during a four-week period in April. Providers receive a spreadsheet preloaded with names, ID numbers and contract numbers for the individuals they support. For each individual, the provider enters cumulative information for the four-week period. Historically, this information was then loaded onto a CD and sent to the ID/DD agency. For the FY2009 data collection, a web system was developed to allow providers to report data directly to the ICI through a secure web interface. Providers are required to submit this annual data as part of their provider contracts. Initially the ID/DD agency was collecting data twice per year, but then subsequently moved to an annual collection to reduce the burden on providers and the Department.

The ID/DD agency passes the data on to staff at the Institute for Community Inclusion, their local University Center of Excellence in Developmental Disabilities, UCEDD, who manage and clean the data and flag issues for follow-up. Follow up and clarification of the data with providers is done by ID/DD agency staff.

New Hampshire. In New Hampshire, employment providers must submit the data to their Area Agency (10 throughout the state), who enter the data into the statewide employment database, and export the data to the state ID/DD agency. For individuals who direct their own services employment data is completed by case managers who manage their service funds. Data is collected on all jobs in New Hampshire, including those in sheltered workshops.

The first time employment data is collected on an individual both the demographic and employment sections of the survey must be completed. For subsequent collection periods Area

Agencies print out the last employment survey for each individual and ask provider staff to update the data. The data reported on hours worked and wages earned is reported as an average for the six month period. After all necessary changes have been made, the employment surveys are returned to the Area Agency. When employment vendors supply data that looks inaccurate or incomplete, Area Agency staff contact the providers for clarification prior to exporting the data to the ID/DD agency. Area Agencies have 40 days after the end of the data reporting period to export their data. To ensure that the exported data remains confidential, the ID/DD agency assigns each individual a code number. Once the exported data is received, the system is automatically linked to the coded individual database at the ID/DD agency's central office.

Washington. In Washington, state counties are responsible for contracting with vendors to provide day and employment services. The data collection system is an integral part of the billing reporting process that vendors, counties, and the ID/DD agency engage in to fund services. Vendors provide outcome data on the activities for each individual for the billing month. The vendor or a county ID/DD agency staff member enters the data into an excel spreadsheet, which is then uploaded to the state ID/DD office. This data is used by the state ID/DD agency to reimburse counties for the services that they have paid vendors to provide.

Data shapes relationships with providers

The data collection process helps providers and the state ID/DD agency to connect in many ways around the goals of employment. The actual implementation of the data collection process establishes a regular and consistent dialogue between providers and the ID/DD agency, and supports providers to understand and work toward the shared goal of increased employment.

Frequency and consistency of data collection helps to cement relationships. Frequent data collection in Florida was seen as a tool to foster relationships with providers and keep them familiar with the data collection process. Having one point person, a Supported Employment liaison, consistently collect data on a monthly basis from providers lets them know that it is a top priority for the Florida's ID/DD agency, so that providers and the ID/DD agency can have a shared understanding about the importance of data in achieving the system's goals. In addition, regular interaction with providers helps to give the SE liaison a month-to-month compass point on the progress providers are making toward their goals.

Outcome data helps providers to become more in tune with their role in increasing employment outcomes. Having conversations with providers about the data is key in helping them to establish a more proactive role in a state's employment agenda. These conversations were taking place during the data collection period, such as was the case in Florida, but more likely they were occurring after the data had been analyzed and was being presented back to providers.

In one local area in Florida, a Supported Employment liaison felt the monthly data collection contact she has with providers is a direct entrée to talk about questions or concerns regarding progress toward employment goals. She uses the consistent check-ins as built-in opportunities for training and technical assistance in specific areas. These conversations can equip providers with the tools they need to increase their employment outcomes.

In Washington, several counties have used their systems' data to identify weaknesses within their contracted provider agencies and develop plans to correct the problems; and as a way to supply feedback to providers about the amount and types of employment outcomes they have provided. A county noted that they have used the data to have conversations with providers about the relationship between services and outcomes. One respondent felt that providers appreciated that the data was being shared with them, reinforcing the sense of ownership providers feel about the types of outcomes they produce. Washington is the only state where, in some counties, they share provider-level data among providers, creating a sense of awareness and competition among providers. Other states do it at a regional or county level

Going a step further, other states are using the data for more direct accountability from providers. In Massachusetts, when the ID/DD agency engages in contract renewal with providers, the expectation is that they use the data to assess outcomes and progress toward identified goals and to set goals for the upcoming contract year. A provider noted his agency makes good use of the data during contract negotiations and goal setting. While use of data for goal setting seems to be a priority in some areas, there is variance among the local area agencies across the state. Another provider noted that they did not currently have targets in their contracts but that they were moving in that direction for the future.

Designing a system that conveys the goal

While the four states examined for this study developed data collection systems to meet their states' unique needs, they each underwent similar processes in the development of the basic components of their systems. The first steps each state took were to identify the frequency of their data collection, the population to collect data on, employment variables, responsibility for collection of data, and responsibility for analysis of the data (please see Table 13). Most important to consider about each of these components is how they help the state further their goals around employment. Each of the elements influences how effective the system is in gathering the data sought and conveying the message of the importance of employment.

Frequency of data collection. As noted, one of the defining elements of Florida's system is its monthly data collection. Respondents spoke of this frequency for the most part as a boon for several reasons, from relationship-development with providers, to ease of reporting because of the frequency of engagement. Washington's data collection system is tied to its service billing system and therefore coincides with the monthly billing cycle. While the frequency of the data collection was seen as an indicator of its significance (said one Supported Employment liaison in Florida, "If it wasn't important, why would I chase them down all month and not let go until I get what I need?"), other states such as Massachusetts and New Hampshire collect data far less frequently (once per year and twice per year, respectively) and feel this is sufficient to assess their state's progress over time.

Targets for data collection. With the exception of Florida, the three other states collect information on all individuals who are working either in the community or in facility-based services (sheltered workshops). Florida only collects information on individuals in community jobs. The types of work activities that are "counted" can be seen as important because it is another element that reflects states' priorities. Other states, such as New Hampshire and

Washington, however, keep track of the number of individuals in sheltered workshops so they can ensure that this number is declining or at least showing little to no growth.

Employment variables. All four states collect information on the type of job (group supported employment, individual employment, etc.), or the amount of time spent in each service category or activity, and wages and hours worked. Some states, like Florida and New Hampshire, collect information on other variables such as receipt of Vocational Rehabilitation services, use of assistive technology, or career advancement opportunities. While data such as these are valuable in assessing the quality of the overall work situation, they are often difficult to obtain.

Massachusetts takes a straightforward approach and only collects information they see as most relevant to understanding who is employed and who is not, and who has entered into a new job during the data collection period. This variable has been added recently and speaks directly to their focus on outcomes.

Responsibility for data collection and analysis. In all cases, employment data is gathered locally at the provider level and submitted to the state ID/DD agency, either at the state level or to a regional office level within the state. With the exception of Massachusetts, all analysis is conducted within the state agency. Massachusetts has contracted with the ICI to conduct the analysis and provide the information back to the state for its use with its regional offices, and providers.

Table 3. Basic elements of state data collection systems

State	Frequency of data collection	Targeted Population	Employment variables	Responsibility for data collection	Responsibility for data analysis
FL	Monthly	All individuals receiving services from the ID/DD agency who are working in the community, and individuals who are eligible to receive services (on the waiting list) who are working	Employer, job title, wages, average hours worked per week, date of last raise, career advancement opportunities, employer-based benefits, integrated work setting, funding source, VR referral information	Employment data is gathered locally. Supported Employment liaisons in each region are responsible for data collection.	Analysis is done within the Employment Unit of the state ID/DD agency
MA	Once annually during a four-week period	All individuals supported by the ID/DD agency employment contracts. Does not include day habilitation (funded through Medicaid state plan) or Community Based	Hours spent in each activity: individual and group SE, facility-based employment, volunteer work, other non-paid day services, and “in transition;” total gross wages; whether	Providers are required to submit their data to the ID/DD agency.	ID/DD agency contracts with the Institute for Community Inclusion to produce employment reports

State	Frequency of data collection	Targeted Population	Employment variables	Responsibility for data collection	Responsibility for data analysis
		Day Services	the individual was employed for 10 of last 12 months in individual or group SE; whether the individual entered a new individual job during previous 12 months		
NH	Twice annually	All individuals receiving services who have had paid employment in the community or a sheltered workshop during the last six months and on each job an individual obtained or maintained during that period of time	Individual demographics, area agency, provider, employer, job tenure, hours worked, wages, work environment, employment benefits, employer incentives, assistive technology, transportation, vocational rehabilitation services	Providers submit data to the Area Agency in their region. The Area Agency enters the data into the statewide employment database, and exports the data to the state.	State ID/DD agency
WA	Monthly	All individuals receiving day and employment services	Data is directly linked to the billing system. Within each service category (individual, group, and prevocational employment; person-to-person; and community access services), data is collected on individual demographics, provider, funding sources, service start date, client hours (paid, volunteer, or other), gross wages, provider service units delivered and cost of service	The vendor or a county ID/DD staff member enters outcome data into an Excel spreadsheet which is then uploaded by the county to the state ID/DD office.	The state ID/DD agency produces standardized reporting measures. The analysis is reported by region, county and provider.

States' use of the data

The four state data collection systems studied used the data they collected at both the state and local levels. While state level data helped states to understand progress toward employment goals, local level data provided information that allowed the ID/DD agency and providers to develop objectives for technical assistance and also has created a sense of awareness among providers with respect to how other providers are doing. Local level administrators (e.g., counties and area agencies) are using the data to have conversations with providers about the relationship between services and outcomes. It should be noted, though, that there are differences between and within states as to how the collected data is used.

State level use of data. States are using this level of data to understand trends at the most macro-level for their agency. State level reports have allowed New Hampshire to see that, while over time more people are getting jobs, the rate of job attainment has not kept pace with growth in the number of people served due to the movement of individuals off the agency's waiting list and transitioning from school to adult life. The reports challenged New Hampshire to do a better job of expanding opportunities for integrated employment. The reports showed that only about 700 businesses in the state employed people supported by the Bureau, despite the fact that New Hampshire has approximately 30,000 businesses.

In Massachusetts, statewide and regional summaries of employment trends are produced. DDS shares these reports, including a three-year historical analysis of employment outcomes in Massachusetts from 2004–2006, with stakeholders. In addition to printed copies of the reports, the summary data are available to DDS staff on demand from a web-based data system developed at ICI. The web-based system allows users to produce graphs or summary reports for each of the variables at the state, regional, or provider levels. Data access is currently restricted to state staff using a password. Custom presentations have also been developed for review by the regional quality improvement teams.

Florida also uses the internet to highlight data. Line graphs documenting the state's progress in meeting its employment goals are posted on its public website. On its intranet, area offices have access to a table that provides employment data for each area. Additionally, in Florida reports are produced on a monthly basis and are shared with each region and staff within ID/DD central office.

In Washington, data are used to complete standard reports including cost/benefit ratios (ratio of service cost to income earned) broken down by region, county, and provider. Data reports have a focus on total number of individuals receiving services, total and average wages, total and average hours worked, total and average state ID/DD agency funding share, and total and average service hours or units.

Local level use of data. Several states reported using data at the local level to motivate providers. Staff at one area agency in New Hampshire noted they use the data to see if their region is producing better employment outcomes than other areas in the state. In Massachusetts, individual provider reports are shared with providers and contain comparisons to regional and statewide averages for placement rates. Florida's data is analyzed by region, and each region can view its own report and generate state-wide reports. While regional office staff have provider-level data,

available data reports do not compare one provider to another across the region or across the state. At the county level in Washington, reports have been used to encourage providers to focus on employment and to increase understanding about trends in employment outcomes. Counties can access the data system and retrieve data. Some counties also have customized software that gave them more flexibility to work with the data, while others maintain their own data systems to retain outcomes information from the billing data and use this data as part of the evaluation of providers' services.

Data are also used to facilitate technical assistance to providers at multiple stages. In Massachusetts, the state ID/DD Agency uses its individual-level provider reports to facilitate conversations with providers about their performance and areas where providers would benefit from support. The conversations often result in the provision of technical assistance to address issues related to employment outcomes. As noted earlier, this is in contrast to Florida, where at times the provision of technical assistance happens at an earlier stage, when supported employment liaisons are collecting the data from providers.

Local level data is not only being used to measure performance, but to set performance expectations. For FY2009, each Massachusetts ID/DD agency area office is using local data to establish a goal for improving employment outcomes and was required to identify and implement at least one strategy to support change. Strategies were varied and included focusing on transition-age young adults, engaging families and residential staff, collaborating with other agencies and resources, building local capacity through training and technical assistance, engaging in strategic planning, and working with employers.

Challenges in developing and implementing a data system

States described challenges that were encountered when they were beginning to implement their systems and ongoing implementation problems with collecting, maintaining, and using the data.

- **Buy-in from providers.** States struggled initially with ensuring that providers were onboard with the data collection initiative and that providers knew how to report and were aware of the importance of timely and accurate reporting. Some providers lacked a mechanism to keep track of outcomes initially.
- **State vs. local control.** In a system like New Hampshire, where local control is institutionalized, standardization across the state of the data reporting system was initially difficult but viewed as critical in maintaining the integrity of the data.
- **Turnover.** Turnover at the provider level continues to affect states' ability to provide good data. Turnover is also a problem for support coordinators, who are additional data reporters, in the Florida ID/DD system.
- **Collaboration with VR.** Several state ID/DD agencies said that they had difficulty working with the state VR office to track individuals who were receiving VR funding for job development and stabilization.
- **Refinement of variables.** Data elements such as wage and hours, while important, do not tell the whole employment story. States report that variables that assess quality of life issues are necessary, though much more difficult to collect.
- **Use of employment data to evaluate providers.** While Massachusetts uses its employment data as part of renegotiating provider contracts, New Hampshire's ID/DD

agency can only use the data to evaluate area agencies and not specific providers.

- **Requirements for reporting data.** In states such as Massachusetts and Washington, where data is tied to contracting and billing, providers have a built-in incentive to provide data. Florida is considering making data reporting a requirement in provider contracts. New Hampshire has not made data a formal requirement, but it is a task that has been ingrained in state ID/DD administrative practices.
- **Access to data.** Overall, individuals and families do not have good access to data reports (although Florida does post its progress toward its 5 Year Initiative on its website).

IMPLICATIONS

Data is important from both a strategic and program planning perspective, and these states are at various stages in using data to drive policy and practice-level decision-making. Respondents reflected that the data collection process as well as the data it produces has been critical in informing and promoting conversation about employment. In some of these states, data has not only been used at local office levels to establish priorities and goals but can also support decision making at the provider level as they receive current data about performance in relationship to local areas and their state. Notwithstanding the challenges noted above, states described several recommendations that can be used by other states as they move toward more comprehensive data collection systems that help to promote their goals.

Goal setting

- Be clear about the goals of data collection initiatives and its link to the overarching goal around improving employment outcomes.
- Be consistent across the state with respect to how area offices are using data for goal setting with providers.
- Set goals for employment outcomes and design a data collection system to measure the employment outcomes valued by your state
- Keep it simple—focus on priority outcomes and be careful not to add in too many process variables.

Establishing a clear message about data collection

- Create policy language that makes it a requirement for supported employment providers and other necessary data sources to provide data.
- The message is clearest when data is collected on all individuals who receive day or employment supports, rather than only targeting specific individuals or contracts.
- Ensure a clear plan for using the data after it is collected. Consider how to use it in evaluation and goal setting, and develop a method for sharing the information with all relevant stakeholders.

Communicating across the system

- Ensure that different areas and regions have opportunities to share one another's priorities and activities around employment.
- Share the data with providers and key stakeholders. States may also want to consider making provider performance information available to individuals and family members to support the selection of providers.

- Ensure that communication reflects data as a priority at the state, regional, and local levels.

Strategies for implementation

- Understand the state agency's culture and develop the system around these parameters.
- Recognize that it takes time to develop a reliable and accurate reporting system. Build time for piloting into the development period and plan on revising the system on an ongoing basis until it functions efficiently for all stakeholders.
- Involve stakeholders such as providers in the design of the system, making sure to include individuals who are knowledgeable about the state's integrated employment services.
- Understand that it is often necessary to involve multiple sources for a complete picture of an individual's employment situation.
- Consider the development of a secure online system to reduce the burden on those responsible for reporting data.

CONCLUSION

With a growing emphasis on accountability, states are beginning to use their data collection systems as part of an overall plan to communicate about and facilitate their states' progress toward greater integrated employment. Through investigation of the important elements in designing and using a data collection system, the variables that provide the most utility for influencing policy, and how data has been used as a strategic planning tool, this chapter provides a framework, although not a blueprint, for how states can use data to effectively communicate their priorities. While most states are at various stages in the process of refining their systems, it is clear that the use of data is one of the most important tools state ID/DD agencies can use to develop, implement, and evaluate their goals.

Based on case study data and recommendations from stakeholders about the most important elements in designing such systems, we suggest some federal guidance for states as they work to develop effective strategies. Guidelines could be based on best practices identified through careful study of successful systems, such as those presented in this chapter and through further research, and help states to work toward establishing national standards for data collection and reporting of employment outcomes. This would be beneficial not only from a state perspective as they create and refine their systems, but would also help to standardize national data comparisons, such as those available through ICI's National Survey of Day and Employment Programs.

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